

# Strings

# String Basics

- A string is a sequence of characters that represents textual data like a person's name, a location, or a message to the user.
- It can be taken as input or be defined in a programs code
- A string literal is defined in the source code directly using single or double quotes

# String Basics

- Sequence type
  - Each element has a position
- immutable
- Holds alpha-numeric characters and symbols
- Has a length

# Indexing

- Consider the following string "ABCDEFGH"
- it has a **length** of 7 elements
  - number of elements in the sequence
- Each element can be individually accessed by its index

# Indexing

- the index indicates its position in the sequence
- the first index value is 0

```
alpha = "ABCDEFGFG"  
print(alpha[0])
```

# Indexing

| Element | A | B | C | D | E | F | G |
|---------|---|---|---|---|---|---|---|
| Index   | 0 | 1 | 2 | 3 | 4 | 5 | 6 |

# Indexing

- Negative indexing is possible

| Element | A  | B  | C  | D  | E  | F  | G  |
|---------|----|----|----|----|----|----|----|
| Index   | -7 | -6 | -5 | -4 | -3 | -2 | -1 |

# Slicing

- To get a **substring** of a given string, we can use slicing
- `string[start:stop]`
  - Indexes must be integers

| Element   | A     | B     | C | D | E    | F    | G | -    | Slice  | Result    |
|-----------|-------|-------|---|---|------|------|---|------|--------|-----------|
| Index     | 0     | 1     | 2 | 3 | 4    | 5    | 6 | 7    | -      | -         |
| Start:End |       | Start |   |   | Stop |      |   |      | [1: 4] | "BCD"     |
| Start:End | Start |       |   |   |      | Stop |   |      | [0: 5] | "ABCDE"   |
| Start:End | Start |       |   |   |      |      |   | Stop | [0: 7] | "ABCDEFG" |

# String Methods

| Method                                                         | Explanation                                                                                                                                    | Result    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <code>alpha.length()</code>                                    | returns the length of alpha                                                                                                                    | 7         |
| <code>alpha.replace('A', 'Z')</code>                           | returns a copy of a string with the given substring replaced with the replacement string.                                                      | "ZBCDEFG" |
| <code>alpha.find("D")</code>                                   | returns the index of the first occurrence of the value given. If the value does not occur, the value returned is -1.                           | 3         |
| <code>alpha.find("X", 2)</code>                                | returns the index of the first occurrence of the value given starting at a given index. If the value does not occur, the value returned is -1. | -1        |
| <code>alpha.find("B", 0, 3)</code>                             | returns the index of the first occurrence of the value given between the starting and end point. Returns -1 if not found.                      | 1         |
| <code>alpha.count('A')</code><br><code>alpha.count('X')</code> | Returns the number of times a given substring appears in the string.                                                                           | 1<br>0    |

# String Equality Methods

| Method        | Explanation                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------|
| isalnum()     | Returns True if all characters in the string are lower case or upper case letters or numbers  |
| isdigit()     | Returns True if all characters are the numbers 0-9                                            |
| islower()     | Returns True if all cased characters are lowercase letters. Will ignore non-cased characters. |
| isupper()     | Returns True if all cased characters are uppercase letters. Will ignore non-cased characters. |
| isspace()     | Returns True if all characters are whitespace.                                                |
| startswith(x) | Returns true if the string starts with x.                                                     |
| endswith(x)   | Returns True if the string ends with x.                                                       |